

Do components correspond to cross-layer virtual attention heads?

A virtual head is a PAIR of heads whose circuits multiply through the residual stream. Scoring that (Elhage et al.) recovers the textbook previous-token $\rightarrow$ induction circuit at $2.05\times$ chance, and specifically on K-composition — so the measure is sound. Head pairs that share a component nonetheless compose at chance. What components DO capture is heads playing the same role in different layers: 360 of 4096 are significantly induction-enriched, and co-owned cross-layer pairs are 14% closer in behaviour than random pairs.

